

ECON 0150 | Economic Data

The economist's data analysis s

Part 2.2 | Numerical Variables by

Behavioral Response to Incentives

How do buyers respond to different discount structures?

- *Starbucks sent different promotional offers to different*
- *Each offer has a different structure (BOGO, \$2 off \$1*

Question: *Which incentive structure affects buying beha*

The Data

Let's load the data and take a look

	Event	Revenue	Offer
0	transaction	34.56	2off1
1	transaction	18.97	2off1
2	transaction	33.90	Bogo
3	transaction	18.01	Bogo
4	transaction	19.11	Bogo

> *which would we expect customers to respond most to:*

Exercise 2.2 | Revenue by Offer Type

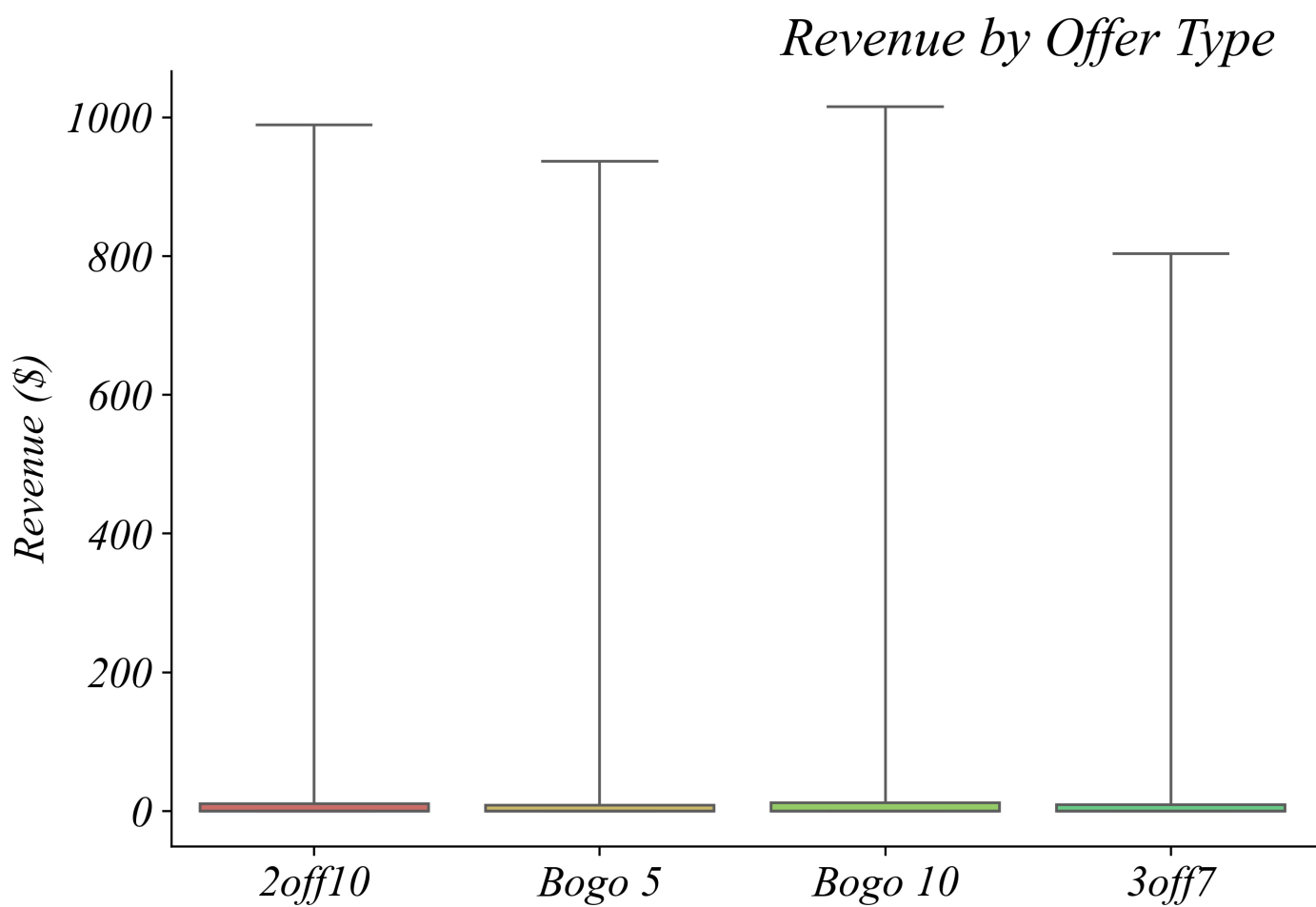
Visualize the data to answer whether Bogo 5 or Bogo 10 has higher revenue

Use a boxplot to show the distribution of numerical variables

```
1 # Boxplot
2 sns.boxplot(data, x='Offer ID', y='Revenue', whis=(0,100))
```

Revenue by Offer Type: Boxplot

The distribution of revenue by offer type.



> *hard to see. Why are so many values compressed at zero?*

Log Transformation: Skewed Data

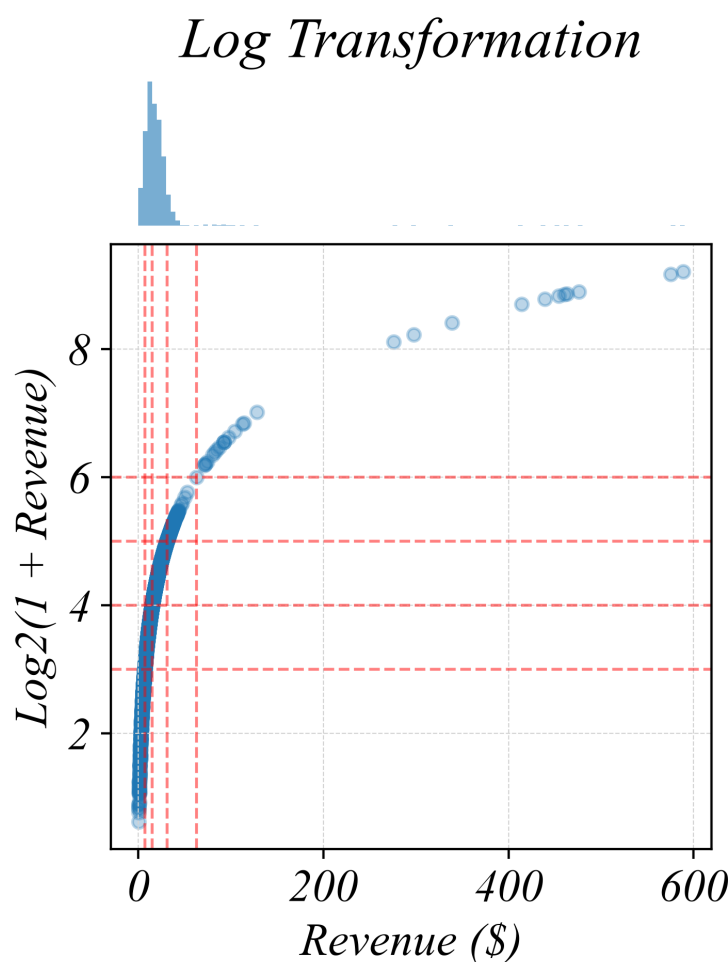
Each unit = a doubling of spending

	Revenue	log2_Revenue
0	34.56	5.152183
1	18.97	4.319762
2	33.90	5.125155
3	18.01	4.248687
4	19.11	4.329841

$> \log_2(1+\$7) = 3, \log_2(1+\$15) = 4, \log_2(1+\$31) = 5$

Log Transformation: Visualized

The transformation spreads out skewed data



> *x-axis is compressed at low values; y-axis spreads the*

Exercise 2.2 | Log Revenue by Offer

Create a boxplot with the log-transformed variable to better see the distribution

Log transform **Revenue**.

```
1 data['log2_Revenue'] = np.log2(1 + data['Revenue'])
```

Create a boxplot of log revenue **log2_Revenue**.

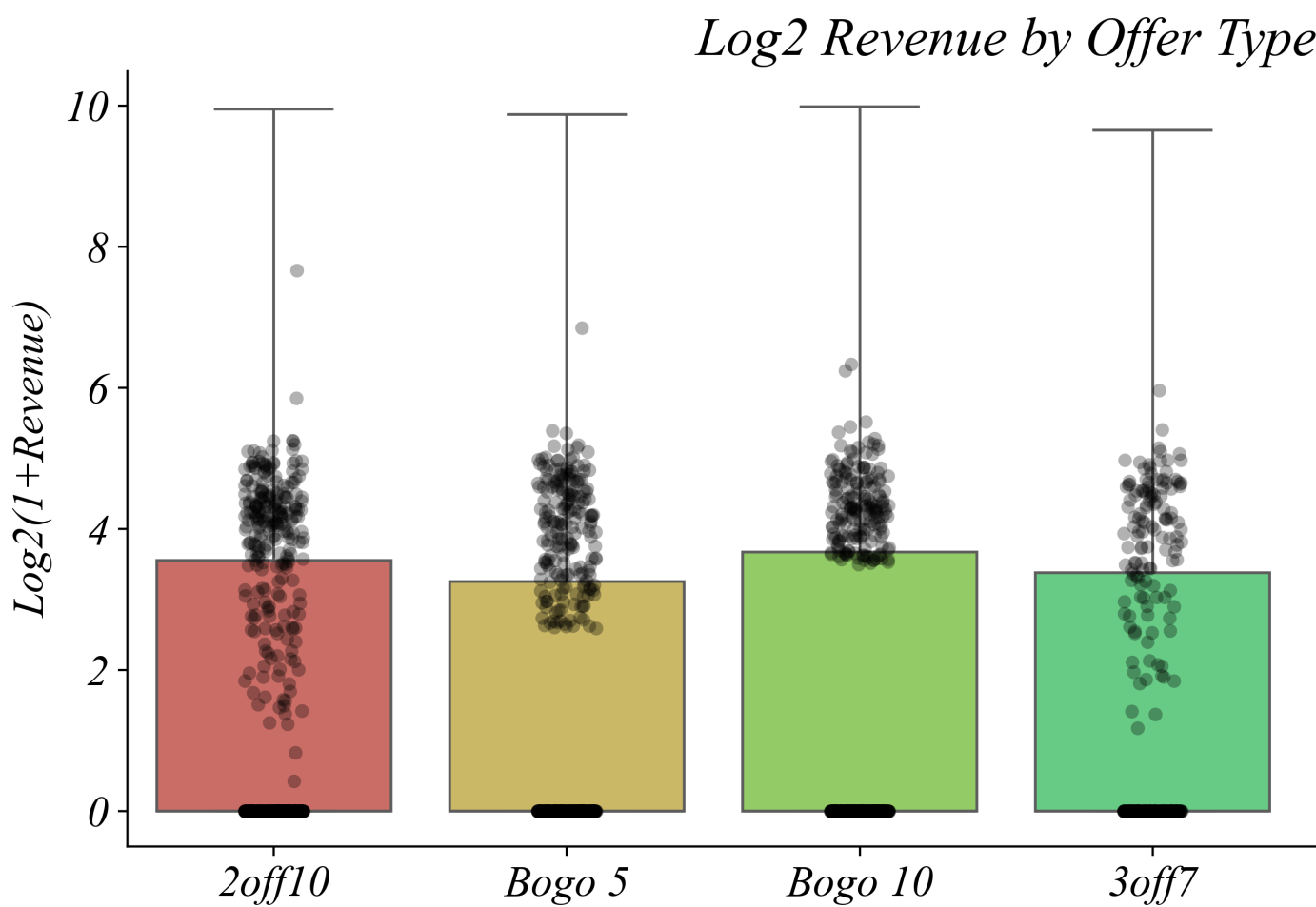
```
1 sns.boxplot(data, x='Offer ID', y='log2_Revenue', whis=(0,100))
```

Add a stripplot.

```
1 sns.stripplot(data, x='Offer ID', y='log2_Revenue', alpha=0.3,
```

Log Revenue by Offer Type: Boxplot

Now we can see the data better.



> *why are there so many zeros?*

Exercise 2.2 | Investigate the Data

What's in the Event column?

Count the unique values in **Event**.

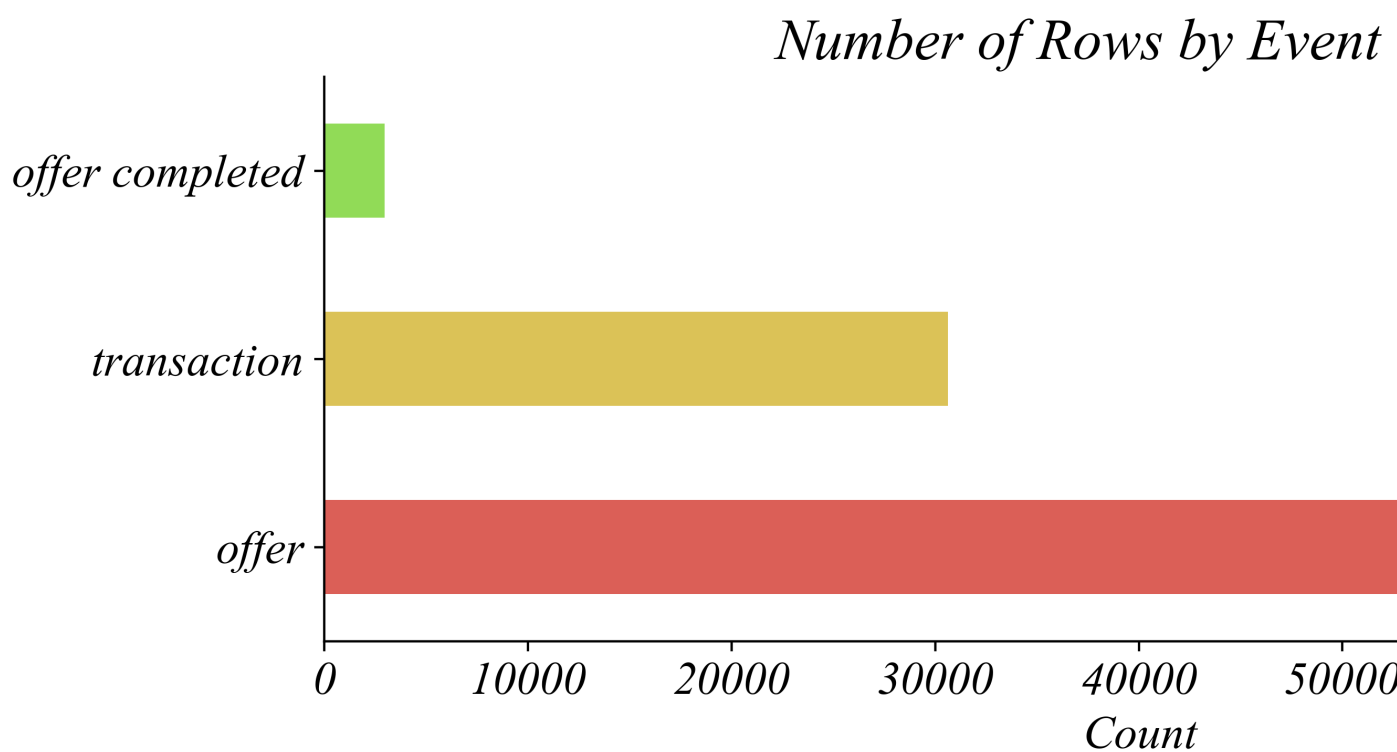
```
1 data['Event'].value_counts()
```

Summarize counts using a countplot.

```
1 sns.countplot(data, x='Event')
```

Three Event Types

Not all rows are purchases



> *most rows are offers, not transactions*

Exercise 2.2 | Revenue by Event

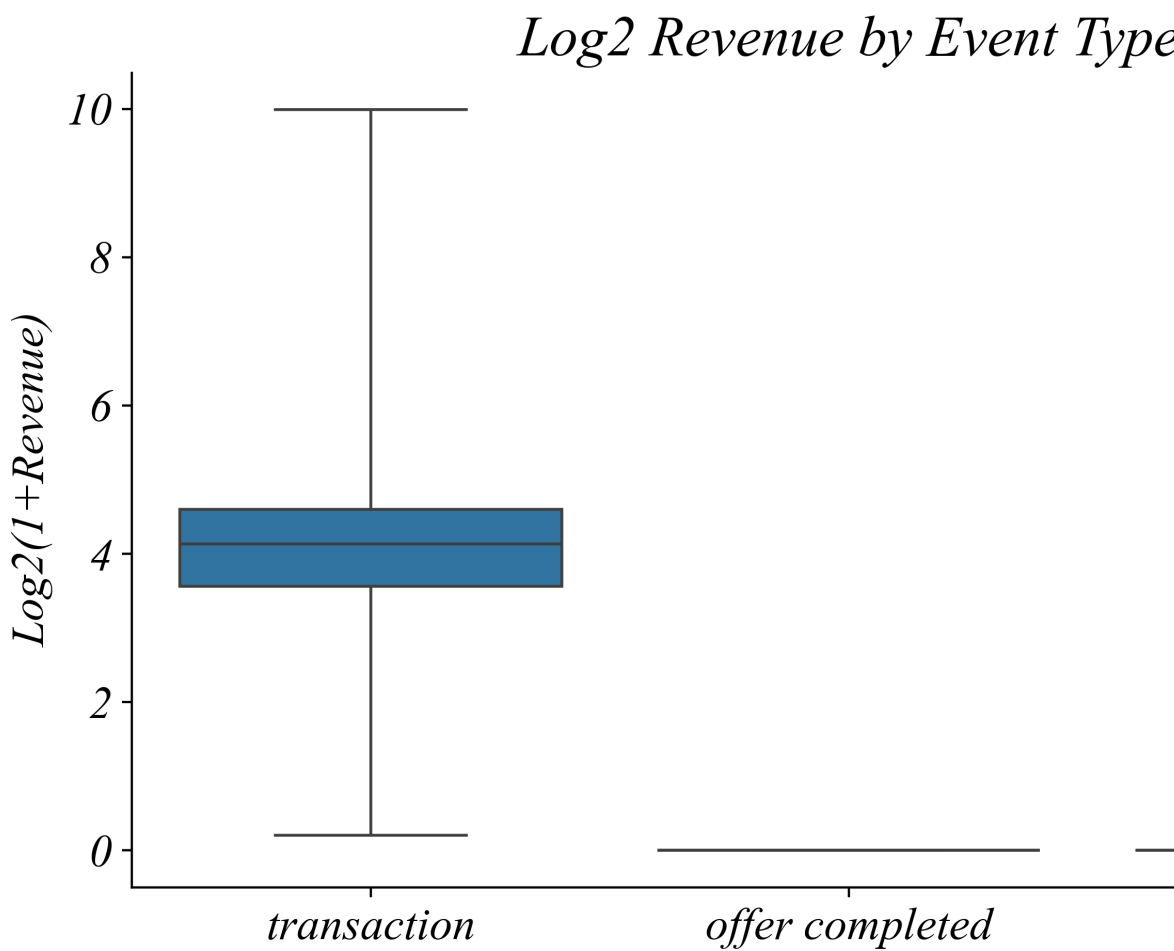
Where is revenue coming from?

Create a boxplot of Revenue by Event.

```
1 sns.boxplot(data, x='Event', y='Revenue', whis=(0,100))
```

Revenue by Event Type

Only transactions have revenue



> offers and completions have zero revenue, which is why

Exercise 2.2 | Summarize Transactions

1. Keep only rows where Event equals `transaction`.

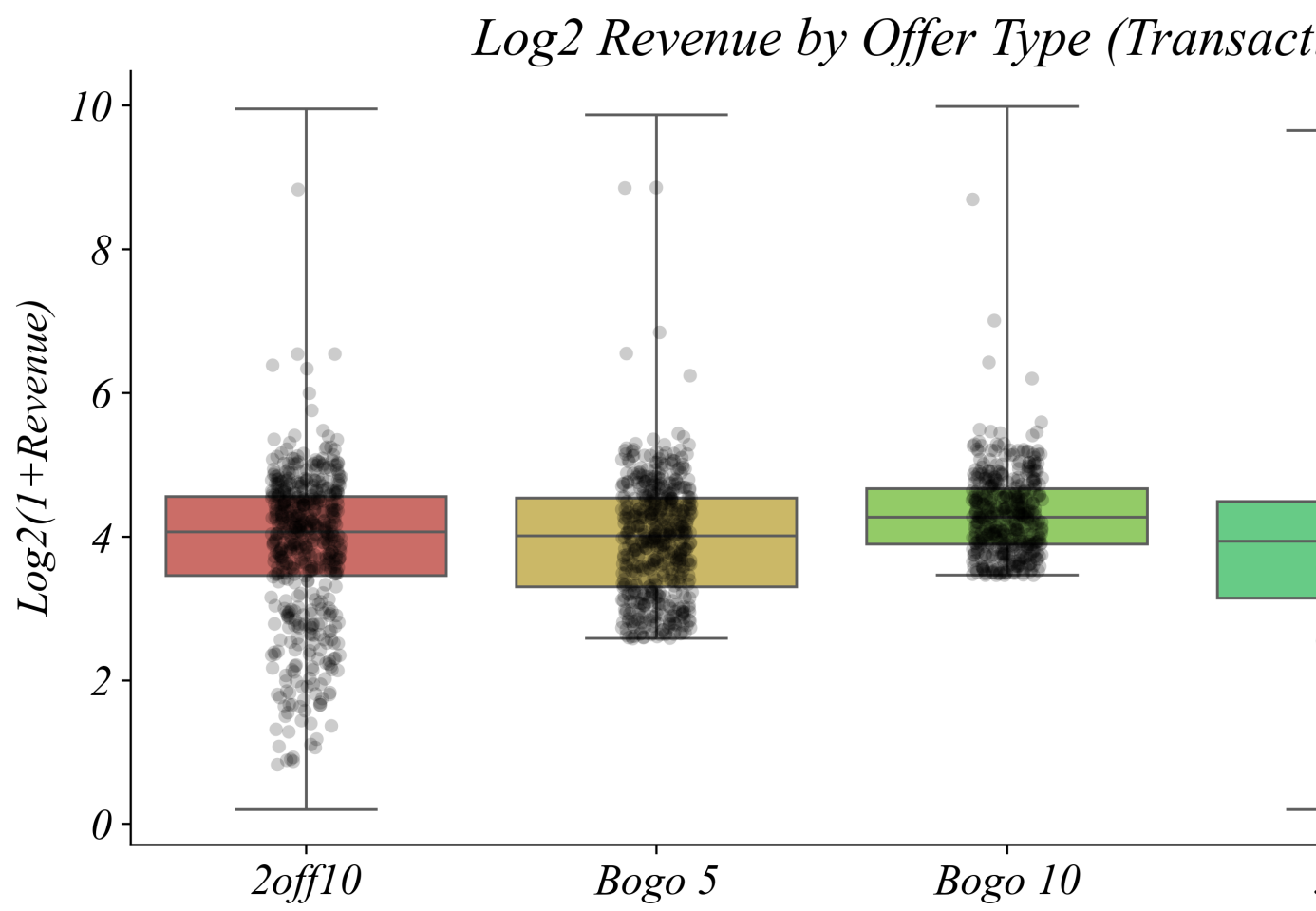
```
1 transactions = data[data['Event'] == 'transaction']
```

2. Create a boxplot of log revenue by offer type using one of the following functions:

```
1 sns.boxplot(transactions, x='Offer ID', y='log2_Revenue', whis=1.5)
2 sns.stripplot(transactions, x='Offer ID', y='log2_Revenue')
```

Summarize Transactions

Every row is a real purchase.



> *which offer type has higher spending?*

Exercise 2.2 | Grouped Statistics

Calculate the mean, standard deviation, and count of log revenue

```
1 transactions.groupby('Offer ID')['log2_Revenue'].agg(['mean', 'std', 'count'])
```

Grouped Statistics

Average log spending by offer type

	mean	std	count
Offer ID			
2off10	3.89	1.03	8569
3off7	3.75	1.06	4698
5off20	4.31	0.89	3239
Bogo 10	4.33	0.65	6308
Bogo 5	3.95	0.81	7803

- > *5off20 has the highest mean*
- > *Bogo 10 has a higher mean than Bogo 5*
- > *but is this the whole story?*

The Workflow

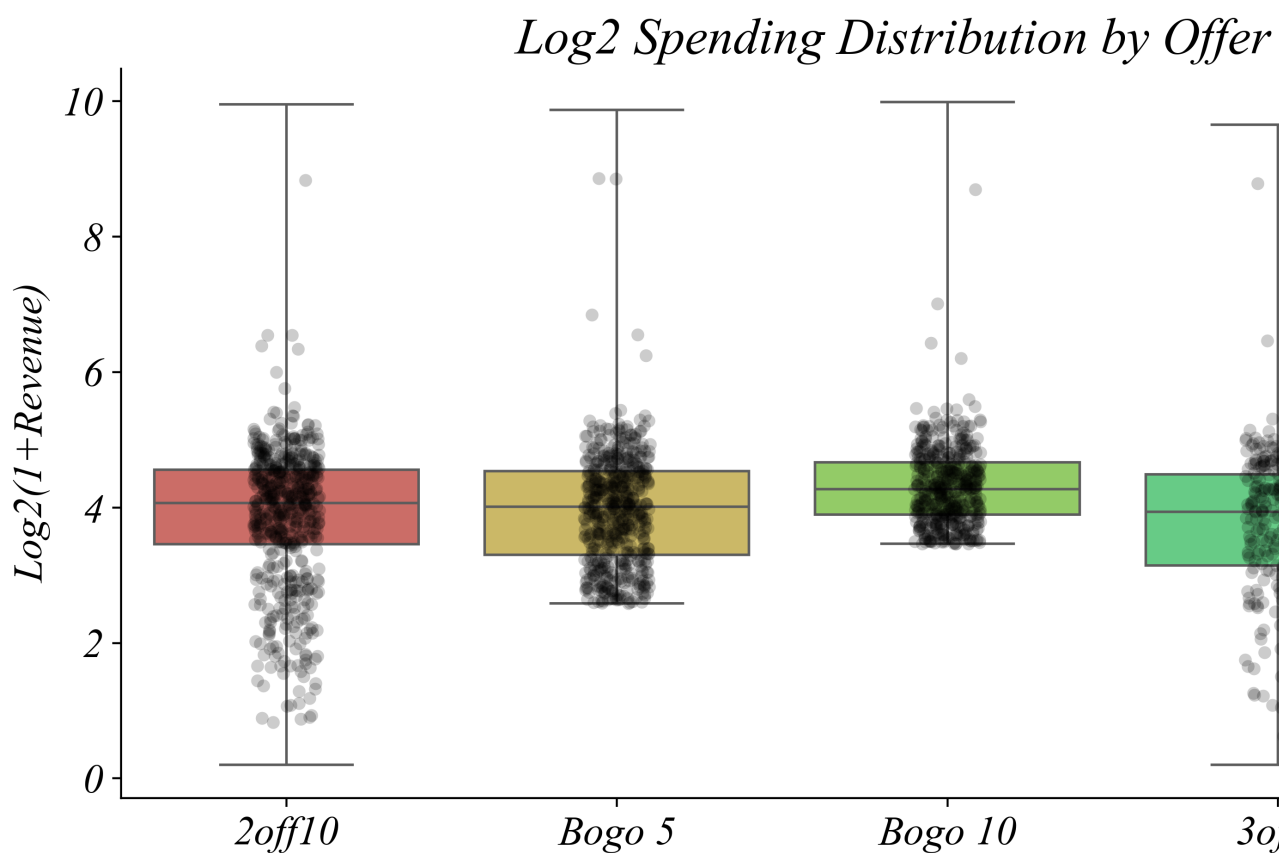
Filter → Transform → Group → Visualize

1. **Filter:** keep only relevant rows
2. **Transform:** log scale for skewed data
3. **Group:** organize by a categorical variable
4. **Summarize:** compare distributions across groups

> you can also see this doesn't always progress in a straight line

Distributions by Offer Type

Each point is one transaction



- > *substantial variation within each offer type*
- > *why are there small purchases in 5off20?*

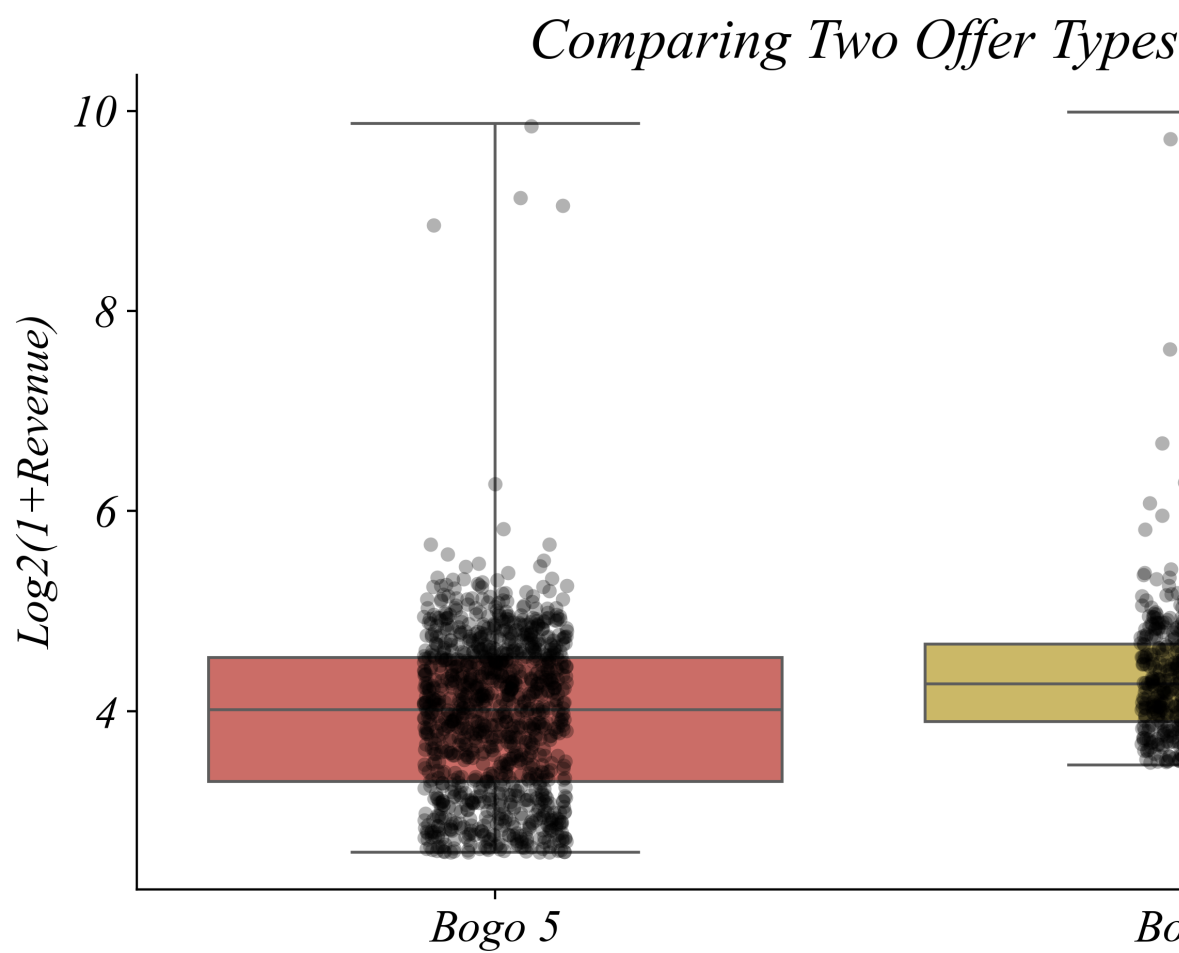
Exercise 2.2 | Compare Two Offers

Filter for just Bogo 5 and Bogo 10, then create a boxplot to compare

```
1 two_offers = transactions[transactions['Offer ID'].isin(['Bogo 5', 'Bogo 10'])]
2 sns.boxplot(two_offers, x='Offer ID', y='log2_Revenue', whis=(0.75, 1.25))
```

Comparing Two Offers

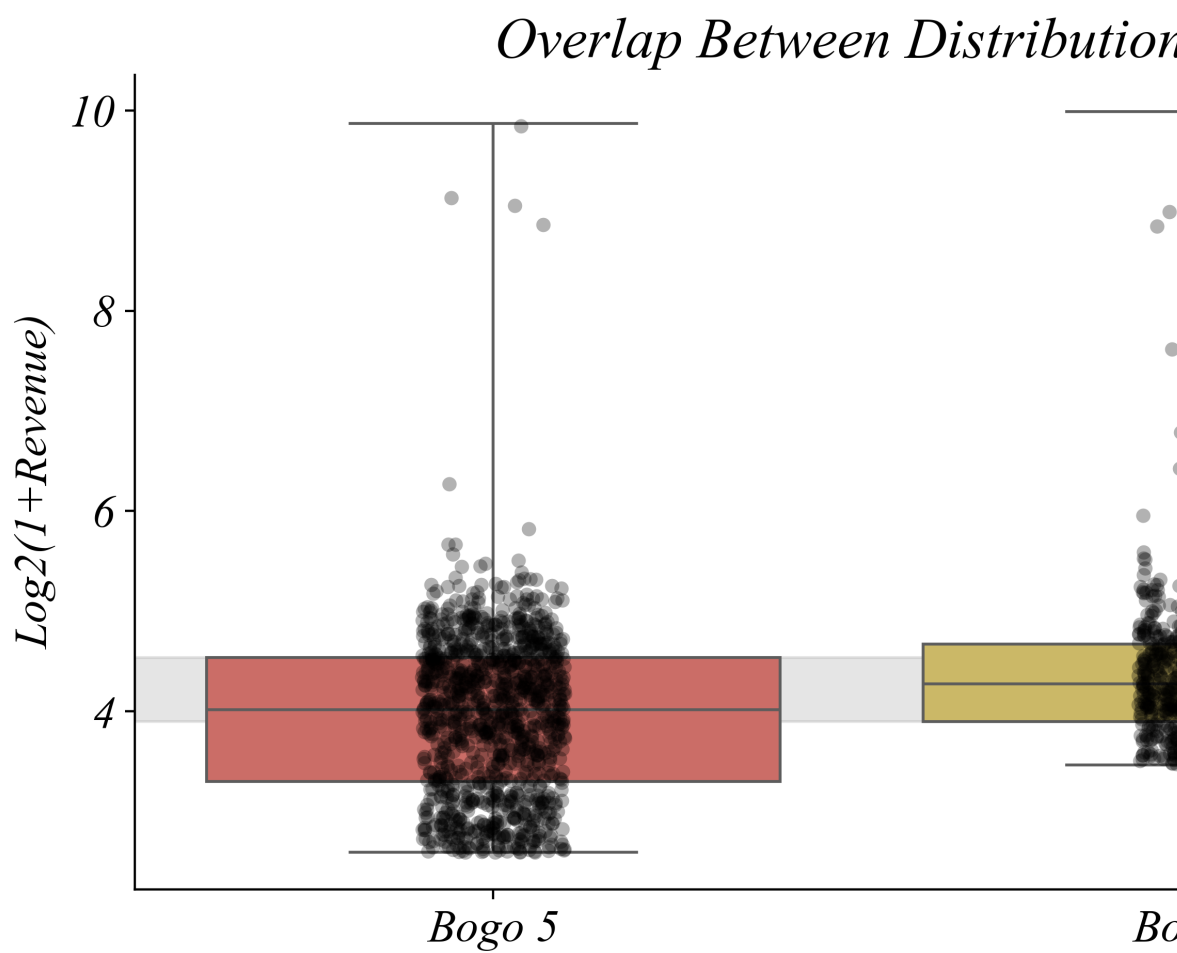
BOGO 5 vs BOGO 10: Do buyers respond differently?



> *BOGO 10 has higher average spending, but look at the*

The Overlap Problem

Many BOGO 5 buyers spent more than BOGO 10 buyers



> when distributions overlap this much, is the difference

The Key Question

Is the difference real or just noise?

- *Average spending differs across offer types*
- *But there's substantial variation within each group*
- *Some “lower” offer buyers outspent “higher” offer buyers*

Question: *Is this difference we observe actually meaningful?*

Part 2.2 | Summary

- *Summary statistics can hide problems: always visualize*
- *Filter your data: make sure you're analyzing what you want*
- *Log transformation helps with skewed data*
- *Boxplots by category show distributions, not just means*
- *Overlapping distributions raise inference questions*

Building Blocks

What this unit adds to your toolkit

Block	Part 2.2
Variables	Numerical + Categorical
Structures	Cross-section
Operations	Filter, Log transform, Group
Visualizations	Bar chart, Boxplot, Stripplo